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THROW LEVER FOR A MAGNIFICATION RING OF AN OPTICAL SIGHT

Abstract

The disclosure is related to a throw lever device for use with at least one magnification ring of at least one optical sight. The throw lever device includes a base member and a lever member pivotally attached to the base member. The lever member includes a detent assembly whereby the lever member may be held in multiple fixed positions about the base member including at least one fixed operable position for turning a magnification ring and at least one fixed non-operable position.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/511,599, filed on Nov. 16, 2023, which is entitled to the benefit of prior-filed U.S. Provisional Patent Application No. 63/426,673, filed on Nov. 18, 2022, the content of which are hereby incorporated by reference in their entirety.

BACKGROUND OF THE DISCLOSURE

1. Field of the Disclosure

[0002] This disclosure relates generally in the field of optical sights for firearms.

2. Background Art

[0003] A conventional variable power optical sight for a firearm includes a magnification ring configured to be manually rotated about an optical axis of the optical sight to change the magnification of the optical sight to a desired magnification. Because some magnification rings are not easily rotatable manually due to poor surface texture and/or positioning along an optical sight, users often attach a raised tab or lever to a magnification ring to assist with manipulating the magnification ring to more quickly and/or more accurately adjust the magnification of the optical sight. One traditional raised lever or “throw lever” includes a cylindrical clamp portion configured to attach to a magnification ring in a manner similar as a pipe clamp and a lever portion extending out from the clamp portion radially from the optical axis of the optical sight. Due to diametrical differences in magnification rings of various optical sights, a user is often required to use a different throw lever for each of his/her specific optical sights.

[0004] Because a traditional throw lever extends out from an optical sight, the throw lever is subject to snagging or catching on items like clothing, user gear, as well as brush and the like when using the optical sight outdoors. To combat this problem, some detachable throw levers have incorporated a hinge in the clamp portion of the throw lever allowing the throw lever to be flipped up and down about the clamp portion. However, cylindrical clamp portions of flip up throw levers are larger or thicker than a traditional clamp portion in order to accommodate a hinge feature. In still other instances, optical sights are manufactured to include a flip up throw lever of a permanent configuration with a spring and detent built into a magnification ring of the optical sight allowing the throw lever to be flipped up and down about the magnification ring. However, a magnification ring of this type is larger than a traditional magnification ring due to the added space required to house the spring and detent. A flip up throw lever is desired that overcomes the above shortcomings.

SUMMARY OF THE DISCLOSURE

[0005] The disclosure provides a device for turning a magnification ring of an optical sight, comprising (1) a base member attachable to the magnification ring; and (2) a lever member pivotally attached to the base member; wherein the lever member has a central longitudinal axis; and wherein the lever member comprises a detent assembly axially disposed about the lever member; and wherein the base member and the detent assembly are operationally configured to retain the lever member in a plurality of fixed positions about the base member.

[0006] The disclosure also provides a system, comprising (1) at least a first optical sight comprising a magnification ring; (2) a device including (a) a base member attachable to the magnification ring, and (b) a lever member pivotally attached to the base member; wherein the lever member has a central longitudinal axis; wherein the lever member comprises a detent assembly axially disposed about the lever member; and wherein the base member and the detent assembly are operationally configured to retain the lever member in a plurality of fixed positions about the base member.

[0007] The disclosure also provides a system, comprising (1) at least a first optical sight and a

second optical sight; (2) at least a first mounting interface and a second mounting interface; (3) a device including (a) a base member; and (b) a lever member pivotally attached to the base member; wherein the first optical sight has a first magnification ring with a first mount surface and wherein the second optical sight has a second magnification ring with a second mount surface; wherein the first mounting interface and the second mounting interface are of a like configuration; wherein the first mounting interface is releasably attachable to the first mount surface and wherein the second mounting interface is releasably attachable to the second mount surface; wherein the base member of the device is releasably attachable to the first mounting interface and releasably attachable to the second mounting interface; wherein the lever member has a central longitudinal axis; and wherein the lever member comprises a detent assembly axially disposed about the lever member.

[0008] The disclosure also provides an assembly for turning a magnification ring of an optical sight, comprising (1) a device including (a) a base member attachable to the magnification ring, and (b) a lever member pivotally attached to the base member; and (2) a mounting interface for the device attachable to the magnification ring; wherein the lever member has a central longitudinal axis; and wherein the lever member comprises a detent assembly axially disposed about the lever member; and wherein the base member and the detent assembly are operationally configured to retain the lever member in a plurality of fixed positions about the base member.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEW OF THE DRAWING

[0009] FIG. 1 is a left side view of an embodiment of a system of this disclosure including an embodiment of a throw lever device secured to an embodiment of a magnification ring of an embodiment of a firearm optical sight via an embodiment of a mounting interface wherein the throw lever device is depicted in a non-operable position.

[0010] FIG. 2 is a perspective view of the system of FIG. 1 depicting the throw lever device set to a non-operable position.

[0011] FIG. 3 is a left side view of the system of FIG. 1 depicting the throw lever device set to an operable position.

[0012] FIG. 4 is a perspective view of the system of FIG. 1 depicting the throw lever device set to an operable position.

[0013] FIG. 5 is a front perspective view of an embodiment of a throw lever device of this disclosure.

[0014] FIG. 6 is a top view of part of a firearm optical sight comprising an embodiment of a magnification ring with a mount surface for the throw lever device of FIG. 5.

[0015] FIG. 7 is a front perspective view of an embodiment of a throw lever device and a mounting interface for use with the throw lever device with the throw lever device depicted in an operable position.

[0016] FIG. 8 is a front perspective view of the throw lever device and the mounting interface of FIG. 7 in an attached configuration with the throw lever device depicted in a non-operable position.

[0017] FIG. 9 is a front perspective view of an embodiment of a throw lever device and a mounting interface for use with the throw lever device with the throw lever device depicted in an operable position.

[0018] FIG. 10 is a front perspective view of the throw lever device and the mounting interface of FIG. 9 in an attached configuration with the throw lever device depicted in a non-operable position.

[0019] FIG. 11 is a left side view of an embodiment of a system of this disclosure including an embodiment of a throw lever device secured to an embodiment of a magnification ring of an embodiment of a firearm optical sight via an embodiment of a mounting interface wherein the throw lever device is depicted in a non-operable position.

[0020] FIG. **12** is a left side view of the system of FIG. **11** depicting the throw lever device set to an operable position.

[0021] FIG. **13** is an exploded view of an embodiment of a throw lever device of this disclosure and an embodiment of a mounting interface for use with the throw lever device.

[0022] FIG. **14** is another exploded view of the mounting interface and throw lever device of FIG. **13**.

[0023] FIG. **15** is a front perspective view of an embodiment of a throw lever device attached to a mounting interface that is mounted to a female dovetail mount surface of a magnification ring of a firearm optical sight depicting the throw lever device set to a non-operable position.

[0024] FIG. **16** is a perspective view of part of the firearm optical sight including the female dovetail mount surface of FIG. **15**.

[0025] FIG. **17** is a top view of part of the firearm optical sight including the female dovetail mount surface of FIG. **15**.

[0026] FIG. **18** is a front perspective view of an embodiment of a base member of a throw lever device of this disclosure.

[0027] FIG. **19** is a rear perspective view of an embodiment of a mounting interface assembled with a throw lever device depicting the throw lever device set to an operable position.

[0028] FIG. **20** is a front perspective view of the mounting interface and throw lever device of FIG. **19** depicting the throw lever device set to a non-operable position.

[0029] FIG. **21** is a front view of the mounting interface and throw lever device of FIG. **19** depicting the throw lever device set to a non-operable position.

[0030] FIG. **22** is a sectional side view of the mounting interface and throw lever device of FIG. **21**.

[0031] FIG. **23** is a rear view of the mounting interface and throw lever device of FIG. **19** depicting the throw lever device set to an operable position.

[0032] FIG. **24** is a sectional side view of the mounting interface and throw lever device of FIG. **23**.

[0033] FIG. **25** is a left side view of the mounting interface and throw lever device of FIG. **19** attached to a magnification ring of a firearm optical sight depicting the throw lever device set to a non-operable position.

[0034] FIG. **26** is a left side view of the mounting interface and throw lever device of FIG. **25** depicting the throw lever device set to an operable position.

[0035] FIG. **27** is a left side view of an embodiment of a mounting interface assembled with a throw lever device depicting the throw lever device set to an operable position.

[0036] FIG. **28** is a left side view depicting the mounting interface and throw lever device of FIG. **27** attached to a female dovetail mount surface of a magnification ring of a firearm optical sight with the throw lever device set to a non-operable position.

[0037] FIG. **29** is a left side view of an embodiment of a mounting interface assembled with a throw lever device depicting the throw lever device set to an operable position.

[0038] FIG. **30** is a perspective rear view of the mounting interface and throw lever device of FIG. **29**.

[0039] FIG. **31** is a top view of the mounting interface and throw lever device of FIG. **29**.

[0040] FIG. **32** is a rear view of the mounting interface and throw lever device of FIG. **29**.

[0041] FIG. **33** is a bottom perspective view of the mounting interface and throw lever device of FIG. **29**.

[0042] FIG. **34** is a right side view of the mounting interface and throw lever device of FIG. **29**.

[0043] FIG. **35** is a bottom view of the mounting interface and throw lever device of FIG. **29**.

[0044] FIG. **36** is a front view of the mounting interface and throw lever device of FIG. **29**.

[0045] FIG. **37** is a front perspective view of the mounting interface and throw lever device of FIG. **29**.

DEFINITIONS USED IN THE DISCLOSURE

[0046] The terms “at least one”, “one or more”, and “one or a plurality” mean one thing or more than one thing with no limit on the exact number; these three terms may be used interchangeably within this disclosure. For example, at least one device means one or more devices or one device and a plurality of devices.

[0047] The term “about” means that a value of a given quantity is within $\pm 20\%$ of the stated value. In other embodiments, the value is within $\pm 15\%$ of the stated value. In other embodiments, the value is within $\pm 10\%$ of the stated value. In other embodiments, the value is within $\pm 7.5\%$ of the stated value. In other embodiments, the value is within $\pm 5\%$ of the stated value. In other embodiments, the value is within $\pm 2.5\%$ of the stated value. In other embodiments, the value is within $\pm 1\%$ of the stated value.

[0048] The term “substantially” or “essentially” means that a value of a given quantity is within $\pm 10\%$ of the stated value. In other embodiments, the value is within $\pm 7.5\%$ of the stated value. In other embodiments, the value is within $\pm 5\%$ of the stated value. In other embodiments, the value is within $\pm 2.5\%$ of the stated value. In other embodiments, the value is within $\pm 1\%$ of the stated value. In other embodiments, the value is within $\pm 0.5\%$ of the stated value. In other embodiments, the value is within $\pm 0.1\%$ of the stated value.

[0049] The term “and/or” includes any and all combinations of one or more of the associated listed items.

DETAILED DESCRIPTION OF THE DISCLOSURE

[0050] For the purposes of promoting an understanding of the principles of the present disclosure, reference is now made to the embodiments illustrated in the drawings and particular language will be used to describe the same. It is understood that no limitation of the scope of the claimed subject matter is intended by way of the present disclosure and the present disclosure is not limited to particular embodiments. As understood by one skilled in the art to which the present disclosure relates, various changes and modifications of the principles as described and illustrated are herein contemplated. It is also to be understood that the terminology used herein is for the purpose of describing particular embodiments only, and is not intended to be limiting. Herein, the term “firearm” may include, but is not limited to a pistol, a semiautomatic firearm, e.g., a semiautomatic rifle, a bolt action firearm, e.g., a bolt action rifle, a shotgun, a revolver, a shoulder fired bazooka, a shoulder fired rocket launcher, an air rifle, and a paintball gun. As understood by the skilled artisan, a particular firearm may be provided in different barrel lengths. Non-limiting examples of pistols are provided in U.S. Pat. No. 4,539,889, titled “Automatic Pistol with Counteracting Spring Control Mechanism,” issued on Sep. 10, 1985; and U.S. Pat. No. D918,328 S, titled “Handgun,” issued on May 4, 2021, each of which is herein incorporated by reference in its entirety. Non-limiting examples of semiautomatic rifles are provided in U.S. Pat. No. 9,777,975 B2, titled “Semiautomatic Firearm,” issued on Oct. 3, 2017; and U.S. Pat. No. 7,775,150 B2, titled “Law Enforcement Carbine with One Piece Receiver,” issued on Aug. 17, 2010, each of which is herein incorporated by reference in its entirety. Non-limiting examples of a bolt action firearm are provided in U.S. Pat. No. 8,925,234 B1, titled “Bolt Action Rifle with Safety Latching Mechanism,” issued on Jan. 6, 2015; and U.S. Pat. No. 8,397,416 B2, titled “Multi-Caliber Bolt-Action Rifle and Components,” issued on Mar. 19, 2013; each of which is herein incorporated by reference in its entirety.

[0051] Herein, a person using a throw lever device of this disclosure may be referred to as a “user” or as a “shooter.” The term “optical sight” may be used interchangeably with the terms “scope,” “scope sight,” “optical platform,” “sight system,” and “telescopic sight.” An optical sight for use with a throw lever device of this disclosure suitably includes at least one turnable member including, but not necessarily limited to a turnable magnification ring and/or a turnable collar common to telescopic sights, spotting telescopic sights, firearm optical sights, and the like. Non-limiting examples of an optical sight for use with a throw lever device of this disclosure include

first focal plane optical sights, second focal plane optical sights, and dual focal plane optical sights. Non-limiting examples of optical sights comprising a magnification ring are described in U.S. Pat. No. 10,180,565 B2 titled "Viewing Optic with an Integrated Display System," issued on Jan. 15, 2019, which is herein incorporated by reference in its entirety; and United States Patent Application Publication Number US 2011/0314720 A1 titled "Rubber Armored Rifle Scope with Integrated External Laser Sight," published on Dec. 29, 2011, which is herein incorporated by reference in its entirety.

[0052] In an embodiment, the disclosure is related to a throw lever device for use with one or more optical sights including, but not limited to one or more firearm variable power optical sights. In an embodiment, the disclosure is related to a low-profile throw lever device for use with a magnification ring of one or more optical sights including, but not limited to one or more firearm variable power optical sights. In an embodiment, a throw lever device may be releasably fastened directly to a magnification ring of an optical sight. In an embodiment, a throw lever device may be releasably fastened to a magnification ring of an optical sight via one or more intermediary members.

[0053] In an embodiment, the disclosure is related to a throw lever device for use with a magnification ring of one or more firearm optical sights, wherein the throw lever device comprises an assembly of components including a base member and a lever member pivotally attached to the base member, the lever member comprising a detent assembly axially disposed about the lever member, wherein the base member and the detent assembly are operationally configured to retain the lever member in a plurality of fixed positions about the base member including a non-operable position and one or more operable positions.

[0054] In another embodiment, the disclosure is related to a system including one or more optical sights and one or more throw lever devices releasably attachable to a turnable magnification ring of each of the one or more optical sights. In an embodiment, the system includes one or more intermediary members for releasably attaching a throw lever device of the one or more throw lever devices to a magnification ring wherein each magnification ring of the one or more optical sights includes a mount surface operationally configured to receive at least a first intermediary member of the one or more intermediary members. Each throw lever device of the one or more throw lever devices includes (1) a base member releasably attachable to the one or more intermediary members and (2) a lever member pivotally attached to the base member, the lever member comprising a detent assembly axially disposed about the lever member, wherein the base member and the detent assembly are operationally configured to retain the lever member in a plurality of fixed positions about the base member.

[0055] In another embodiment, the disclosure is related to a magnification ring throw lever system for one or more optical sights, the system including (1) a throw lever device comprising a base member and a lever member pivotally attached to the base member, and (2) a mounting interface operationally configured to receive the base member in releasable attachment thereto and operationally configured to be releasably attached to one or more magnification rings of one or more firearm optical sights, wherein each magnification ring of the one or more optical sights includes a mount surface for releasable attachment of the mounting interface and wherein the lever member includes a detent assembly axially disposed about the lever member, wherein the base member and the detent assembly are operationally configured to retain the lever member in a plurality of fixed positions about the base member.

[0056] In another embodiment, the disclosure is related to a throw lever device for a magnification ring of a firearm optical sight including a lever member comprising a spring and plunger assembly for holding the lever member in a fixed operable position and a fixed non-operable position.

[0057] In another embodiment, the disclosure is related to a system comprising an interchangeable throw lever device and at least one firearm optical sight comprising a magnification ring, wherein the throw lever device comprises a mounting interface and the magnification ring comprises a

mounting surface for releasable attachment of the mounting interface. In one embodiment, the system includes a plurality of firearm optical sights, each firearm optical sight having a magnification ring with a like mounting surface for receiving the throw lever device in operable attachment.

[0058] In another embodiment, the disclosure is related to a throw lever device for manipulating a turnable ring of a cylindrical member. Although a throw lever device of this disclosure may be built to scale, in one embodiment a throw lever device is sized for manual operation with one or more turnable magnification rings of one or more firearm optical sights.

[0059] In another embodiment, the disclosure is related to a turnable lever device for manipulating a turnable ring of a cylindrical member.

[0060] As described herein, a throw lever or “throw lever device” of this disclosure is operationally configured for use with a magnification ring of an optical sight including, but not limited to a magnification ring of a firearm optical sight. With reference to FIGS. 1-4, when attached to a magnification ring **501** of a firearm optical sight **500** (hereafter “optical sight **500**”), a throw lever device **100** may be directed to a fixed non-operable position as shown in FIGS. 1 and 2 and directed to at least a first fixed operable position as shown in FIGS. 3 and 4. At an operable position, the throw lever device **100** may be used to manually turn a corresponding magnification ring **501** in either direction for purposes of adjusting magnification of an optical sight **500**. As described herein, in an embodiment a throw lever device **100** may be used interchangeably amongst a plurality of optical sights **500**. For example, a throw lever device **100** may be provided as part of a system comprising a plurality of optical sights **500** each having a magnification ring **501** operationally configured to receive a throw lever device **100** in releasable attachment thereto.

[0061] With reference to FIGS. 5 and 6, in an embodiment a throw lever device **100** may comprise a main seat or base member **110** operationally configured as a mounting member of the throw lever device **100** that is releasably attachable to a mount surface **505** of a magnification ring **501** of an optical sight **500**. In this embodiment, the throw lever device **100** further includes a lever member **120** pivotally attached to the base member **110** about a rotational axis in a manner effective to set the lever member **120** to a fixed non-operable position and to a fixed operable position similar as described above in reference to FIGS. 1-4.

[0062] In the embodiment of FIGS. 5 and 6, the base member **110** includes a mounting surface **121** operationally configured to contact or mate with a mount surface **505** of a magnification ring **501** of one or more optical sights **500**. Suitably, the mounting surface **121** comprises a surface configuration operable with the mount surface **505**. For example, the mounting surface **121** may include a planar surface for contacting a planar mount surface **505**. In another embodiment, the mounting surface **121** may include a non-planar surface or other surface configuration and a mount surface **505** may include one or more conformal contacts for the mounting surface **121**. In another embodiment, the mounting surface **121** may include a curved surface having a radius of curvature the same or substantially similar as the radius of curvature of a corresponding magnification ring **501**.

[0063] In the embodiment of FIG. 5, the base member **110** includes one or more through holes **101** corresponding to one or more threaded holes **506** disposed along the mount surface **505** whereby the base member **110** may be releasably fastened to the magnification ring **501** via one or more threaded fasteners **122**. Suitable threaded fasteners **122** may include, but are not limited to socket head cap screws, countersunk screws, flathead screws, and combinations thereof. As such, the one or more through holes **101** may be of a type effective to receive one or more particular types of threaded fasteners **122** there through. For example, a through hole **101** may include a counterbore hole as shown in FIG. 5 for receiving a socket head cap screw **122**. In another non-limiting example, a through hole **101** may include a countersunk hole for receiving a countersunk screw or flathead screw as understood by the skilled artisan. According to the configuration of the base member **110** as described above, in an embodiment a plurality of optical sights **500** may each

comprise a mount surface **505** the same or similar as shown in FIG. **6** whereby the throw lever device **100** may be used as an interchangeable member with each of the optical sights **500**.

[0064] Turning to FIGS. **7-10**, in another embodiment a throw lever device **100** may be releasably attached to a magnification ring **501** of an optical sight **500** via one or more intermediary members such as a mounting interface **105** operationally configured as a uniform attachment for the throw lever device **100** regardless the outer diameter of the magnification ring **501**. In other words, a mounting interface **105** as shown in FIGS. **7-10** provides a mount surface of an unvarying size for a corresponding throw lever device **100** regardless the outer diameter of the magnification ring **501** to which the mounting interface **105** is secured. As such, a single throw lever device **100** may be mounted to different optical sights **500** having magnification rings **501** of different outer diameters. [0065] Similar as described above, a lever member **120** of this embodiment is pivotally attached to the base member **120** and may be oriented or set to a fixed non-operable position and to a fixed operable position for turning a corresponding magnification ring **501**. A throw lever device **100** of this disclosure may include a one piece lever member **120** of a predetermined length, size and shape. In another embodiment, a throw lever device **100** may include two or more interchangeable lever members **120** of varying length and/or size and/or shape. As shown in FIGS. **7-10** and as described below, in another embodiment a lever member **120** may be comprised of an assembly of components operationally configured to establish a lever member **120** of a desired length and/or size and/or shape.

[0066] As shown in FIG. **11**, when a lever member **120** of a throw lever device **100** as shown in FIGS. **7-10** is set to a fixed non-operable position (or “flip down” position), a central longitudinal axis A-A of the lever member **120** is parallel to an optical axis B-B of a corresponding optical sight **500**. As shown in FIG. **12**, when the lever member **120** is set to a fixed operable position (or “flip up” position), the central longitudinal axis A-A is perpendicular to the optical axis B-B wherein the lever member **120** is operationally configured as a radially extending member relative optical axis B-B. In an embodiment, when the throw lever device **100** is set to a fixed operable position the lever member **120** may comprise a length whereby the lever member **120** extends out from a corresponding optical sight **500** a distance greater than any other exterior components of the optical sight **500** as shown in FIG. **12**. In another embodiment, the lever member **120** may include a length whereby one or more other exterior components of a corresponding optical sight **500**, e.g., a turret **502** of the optical sight **500**, may extend out from the optical axis B-B a distance greater than the lever member **120**. In another embodiment, a lever member **120** may extend out from the optical axis B-B a distance the same or substantially similar as one or more other exterior components of a corresponding optical sight **500**. In still another embodiment, a throw lever device **100** may be operationally configured so that the lever member **120** extends out from a corresponding optical sight **500** in a fixed operable position with the central longitudinal axis A-A set at an angle between 0.0 degrees and 90.0 degrees relative optical axis B-B.

[0067] As stated above, although a throw lever device **100** of this disclosure may be built to scale, for manual operation a lever member **120** includes a length and/or size and/or shape suitable to be manipulated by one or more fingers of a hand of a user or other person when secured to a magnification ring **501** of an optical sight **500** whereby force applied to the lever member **120** in either direction applies torque to the corresponding magnification ring **501** for turning the magnification ring **501** in the same direction as the lever member **120**. In one non-limiting embodiment, a distal end of a lever member **120** may extend out from a circumferential surface **503** of a corresponding magnification ring **501** at a distance ranging from or about 2.99 cm to or about 4.34 cm (from or about 1.178 inches to or about 1.71 inches). In addition, a lever member **120** for use with an optical sight **500** may include an outer width or outer diameter ranging from or about 0.88 cm to or about 1.0 cm (from or about 0.346 inches to or about 0.394 inches). Suitably, a force may be applied at any point along the length of the lever member **120** for applying a desired torque to a corresponding magnification ring **501**. In another embodiment, a lever member **120** may be

manually operated to turn a corresponding magnification ring **501** when the lever member **120** is oriented or set to a fixed non-operable position as shown in FIG. **11**.

[0068] As described below, arresting of the lever member **120** in either a fixed non-operable position or a fixed operable position about the base member **110** is achieved according to the configuration of the base member **110** and the lever member **120**. In particular, a lever member **120** of this disclosure includes a detent assembly that is axially disposed about the lever member **120**, i.e., axially disposed about the central longitudinal axis A-A of the lever member **120**, and operationally configured to be biased against different surfaces of the base member **110** to establish different fixed positions of the lever member **120**. By incorporating a detent assembly as part of the lever member **120** the size of a body of a corresponding magnification ring **501** is minimized providing a low profile throw lever device **100**, mounting interface **105** and magnification ring **501** combination.

[0069] With reference to FIGS. **13** and **14**, a throw lever device **100** for use with a mounting interface **105** may comprise an assembly of components including a base member **110** and a lever member **120** pivotally attached thereto. In this embodiment, the lever member **120** comprises an assembly including a post member **125**, an annular plunger **130**, one or more biasing members **135**, an annular boss member **140**, an annular retaining ring **145**, an optional annular first extension member **150** and a threaded fastener, e.g., a set screw **155**, coaxial with longitudinal axis A-A. Herein, the post member **125**, the plunger **130**, the boss member **140** and the one or more biasing members **135** are collectively operationally configured as a detent assembly of the throw lever device **100**. In this embodiment, because the lever member **120** carries the one or more biasing members **135**, the amount of space afforded within the lever member **120** allows for an increase in size and/or stiffness and/or strength of the one or more biasing members **135** compared to a conventional detent requiring a biasing member housed in a magnification ring of an optical sight. For purposes of this disclosure, a detent assembly may also be referred to as a “biased detent assembly.”

[0070] Still referring to FIGS. **13** and **14**, in this embodiment the mounting interface **105** comprises a first male dovetail surface **106** for forming a first dovetail joint with a corresponding slot type female dovetail mount surface **505** of a magnification ring **501** as shown in FIG. **15**. As such, it is herein contemplated that a plurality of optical sights **500** may be provided each having a magnification ring **501** with a female dovetail mount surface **505** as shown in FIGS. **15-17** whereby each mounting interface **105** operates as a universal type mounting interface **105** for a throw lever device **100**. As shown, a base member **110** for use with a mounting interface **105** of this embodiment comprises a slot type female dovetail mount surface **111** for forming a second dovetail joint with a second male dovetail surface **107** of the mounting interface **105** as shown in FIGS. **8**, **10** and **15**. Accordingly, a mounting interface **105** of this embodiment includes two male dovetail connectors or surfaces operationally configured to form two dovetail joints for releasably attaching a throw lever device **100** to a magnification ring **501**.

[0071] As understood by the skilled artisan, a mounting interface **105** of a particular size and shape may determine a minimum circumference magnification ring **501** to be used with the mounting interface **105**. In other words, when using a mounting interface **105** of a particular size and shape, a body of a magnification ring **501** must be large enough to include a corresponding female dovetail mount surface **505** for receiving the mounting interface **105** therein as shown in FIG. **15**. For example, in a non-limiting embodiment including a mounting interface **105** having a height of or about 4.30 mm (0.17 inches) and a first male dovetail connector **106** having an outer width of or about 8.10 mm (0.32 inches), a corresponding a female dovetail mount surface **505** suitably comprises a depth of or about 3.0 mm (0.19 inches) from a circumferential surface **503** of the magnification ring **501** and an inner width for receiving the first male dovetail connector **106** of or about 9.80 mm (0.39 inches). In another embodiment, a circumferential surface **503** of a magnification ring **501** may include additional material forming part of the body of the

magnification ring **501** of a size suitable for forming a female dovetail mount surface **505** therein. [0072] In an embodiment, including a first male dovetail connector **106** and a second male dovetail connector **107** of equal or substantially equal outer widths as depicted, a female dovetail mount surface **111** of a base member **110** suitably includes an inner width the same or substantially similar as the inner width of a female dovetail mount surface **505**. In another embodiment, a first male dovetail connector **106** and a second male dovetail connector **107** of a mounting interface **105** may include dissimilar widths for mating with a female dovetail mount surface **505** of a magnification ring **501** and a female dovetail mount surface **111** of the base member **110** comprising corresponding dissimilar inner widths.

[0073] Referring to FIGS. **13** and **14**, the base member **110** comprises a pivot attachment surface for the lever member **120** provided as a socket type surface (“socket **112**”) operationally configured to receive a proximal end **126** (or “eye end **126**”) of the post member **125** therein for pivoting of the lever member **120**. As shown, the socket **112** includes opposing side wall members **113** and **114** with concentric apertures **115** and **116** (see also FIG. **18**) for receiving a pivot pin **156** there through, wherein the lever member **120** pivots about a rotational axis along a single plane perpendicular to a central axis C-C of the pivot pin **156** (see directional arrow A1 in FIG. **3**). For purposes of this disclosure, the pivot pin **156** may be referred to as a fulcrum of the lever member **120**, i.e., a fulcrum of the throw lever device **100**.

[0074] In this embodiment, the post member **125** comprises a proximal eye end **126** for mating with an inner surface of the socket **112** in a hinge joint configuration. As shown, the eye end **126** includes an aperture **127** for receiving the pivot pin **156** there through when the aperture **127** is concentrically aligned with apertures **115** and **116** of the socket **112**. Suitably, the eye end **126** includes an outer surface configuration suitable for providing angular movement of the post member **125** about the pivot pin **156** within the socket **112** for directing the lever member **120** to a non-operable position as shown in FIGS. **1** and **2** and an operable position as shown in FIGS. **3** and **4**. Without limiting the eye end **126** to a particular configuration, in an embodiment the proximal end of the post member **125** may comprise a curved surface configuration. In an embodiment as shown in FIGS. **13** and **14**, the eye end **126** may comprise a cylindrical type body with a curved surface at a proximal end of the post member **125** with opposing planar outer surfaces (see planar surface **129** in FIG. **13**).

[0075] With further reference to FIGS. **13** and **14**, the plunger **130** (or “detent plunger **130**”) is provided as a ring or band type member with an opening **131** for receiving the post member **125** there through in a manner effective for a proximal end of the plunger **130** to contact part of the socket **112**. In this embodiment, the proximal end of the plunger **130** includes a beveled surface **132** for contacting corresponding mating surfaces of the socket **112**. Referring to FIG. **18**, the mating surfaces of the socket **112** include a first seat **117** of mating surfaces for receiving the beveled surface **132** in abutment thereto when the lever member **120** is set to an operable position as shown in FIG. **19**. As shown in FIG. **19**, the socket **112** includes a second seat **118** perpendicular to the first seat **117** with mating surfaces for receiving the beveled surface **132** in abutment thereto when the lever member **120** is set to a non-operable position as shown in FIG. **20**. As shown in FIGS. **25** and **26**, the configuration of the first seat **117** and the second seat **118** are operationally configured to receive and hold the plunger **130** in a fixed position thereby limiting the range of angular travel of the lever member **120** to or about 90.0 degrees between a non-operable position and an operable position.

[0076] Although the base member **110** of this embodiment includes a first seat **117** and a second seat **118** offset ninety degrees, in another embodiment a base member **110** may include one or more additional seats disposed between the first and second seats **117**, **118** and offset at one or more angles as desired, e.g., a third seat offset at an angle of forty-five degrees. In addition, part of the second seat **118** of the socket **112** may include an extended portion or ledge **119** operationally configured as a stop for the lever member **120** to prevent the lever member **120** from contacting a

corresponding optical sight **500**—see FIG. 25, which depicts the lever member **120** set to a non-operable position against the ledge **119** spaced apart from the optical sight **500**.

[0077] Referring to FIG. 22, the inner surface of the plunger **130** includes a first inner surface **137** of a first inner diameter for receiving the post member **125** there through and a second inner surface **138** of a second inner diameter greater than the first inner diameter providing a space for holding a proximal portion of a biasing member **135** disposed around part of the post member **125** as shown. In this embodiment, a junction between the first inner surface **137** and the second inner surface **138** is defined by an annular seat **133** operationally configured as an abutment surface for a proximal end of the biasing member **135**. In one embodiment, the biasing member **135** may include a helical spring with a proximal end axially disposed within the plunger **130** and a distal end disposed within the boss member **140**. As shown, the boss member **140** is operationally configured as a housing for at least part of the biasing member **135** and at least part of the plunger **130**. In particular, the boss member **140** includes an axial through hole with an open space having a first non-threaded inner surface **139** at a proximal end of the boss member **140** for holding a distal part of the biasing member **135** disposed around part of the post member **125**. As shown, the inner surface **139** includes an inner diameter slightly greater than an outer diameter of the plunger **130** wherein a distal part of the plunger **130** may be disposed or mated within the boss member **140** forming an enclosure of the biasing member **135**. The boss member **140** also includes a first inner threaded surface **141** operationally configured to be threadedly communicated with a distal outer threaded surface **128** of the post member **125** effective for fixing the position of the post member **125** relative the boss member **140**. In this embodiment, a junction between the inner surface **139** and the first inner threaded surface **141** is defined by an annular seat **142** operationally configured as an abutment surface for a distal end of the biasing member **135**. In one mode of assembly, as the boss member **140** is threaded onto the threaded surface **128** of the post member **125**, the boss member **140** is directed linearly toward the base member **110** (see directional arrow A2) whereby the annular seat **142** applies a force to the biasing member **135**, which in turn applies a biasing force to the annular seat **133** effective to bias the plunger **130** against the second seat **118** as shown in FIG. 22 or against the first seat **117** when the lever member **120** is set at an operable position as shown in FIG. 24.

[0078] As shown in FIGS. 7 and 8, the entirety of the lever member **120** of a throw lever device **100** may include the boss member **140** as the most distal component whereby the length of the lever member **120** may be determined, at least in part, according to the length of the boss member **140**. As shown in FIGS. 9 and 10, a lever member **120** may be extended in length via the addition of one or more additional components including at least a first extension member **150** operationally configured to be threadedly communicated with a distal end of the boss member **140**.

[0079] Referring again to FIG. 22, the distal end of the boss member **140** includes an open space defined by an inner surface for receiving a retaining ring **145** therein, the retaining ring **145** having an inner threaded surface **146** operationally configured for threaded communication with the most distal part of the outer threaded surface **128** of the post member **125**. In this embodiment, the open space of the boss member **140** includes a second non-threaded inner surface **147** operationally configured as an abutment surface for part of a proximal end **148** of the retaining ring **145** when the retaining ring **145** is in threaded communication with the post member **125** as shown. In operation, the retaining ring **145** is operationally configured to be adhered in a fixed position and tension the first inner threaded surface **141** of the boss member **140**. Suitable adhesives for use with a retaining ring **145** may include, but are not limited to threadlocker adhesives and two-part epoxy compositions.

[0080] The distal end of the boss member **140** is also operationally configured to receive a proximal end of a first extension member **150** therein in a manner effective to enclose the retaining ring **145** as shown in FIG. 22. In this embodiment, the distal end of the boss member **140** includes a second inner threaded surface **143** for threaded communication with an outer threaded surface

151 of a proximal end of the first extension member **150**. In this embodiment, the first extension member **150** includes an axial through hole including a first inner threaded surface **152** of a first inner diameter at a proximal end of the first extension member **150** for receiving set screw **155** in threaded communication therein. In this embodiment, the set screw **155** is operationally configured to tension the first inner threaded surface **152** of the first extension member **150** when the first extension member **150** is threadedly communicated with the boss member **140**. The first extension member **150** also includes a first non-threaded inner surface **153** of a second inner diameter greater than the first inner diameter providing an open space for accessing the set screw **155** when screwing the set screw **155** on and off of the first inner threaded surface **152** of the first extension member **150**.

[0081] With reference to FIGS. **13** and **14**, the mounting interface **105** may include one or more threaded countersunk holes **108** and **109** for receiving set screws **160** and **161** at a depth therein whereby the proximal ends **160A** and **161A** of the set screws **160** and **161** extend out past a mating surface **164** of the mounting interface **105** (see FIG. **22**) for mating with corresponding female surfaces **507** and **508**, e.g., cavities, dimples, disposed along the female dovetail mount surface **505** of a magnification ring **501** of an optical sight **500** (see FIG. **17**) in a manner effective to load the threads of the set screws **160** and **161** and tension the mounting interface **105** during operation of the throw lever device **100** by forcing the first male dovetail surface **106** against the female dovetail mount surface **505**. In addition, the base member **110** may include one or more threaded holes **165** and **166** for receiving set screws **168** and **169** there through in a manner effective for the proximal ends **168A** and **169A** of the set screws **168** and **169** to mate with the countersunk holes **108** and **109** of the mounting interface **105** (see FIG. **22**). As such, the length of the set screws **160** and **161** is suitably less than the depth of the threaded holes **108** and **109** so that the proximal ends **168A** and **169A** of the larger diameter set screws **168** and **169** are free to mate with the countersunk holes **108** and **109** in a manner effective to load the threads of the set screws **168** and **169** and tension the mounting interface **105** during operation of the throw lever device **100** by forcing the second male dovetail surface **107** against the female dovetail mount surface **111** of the base member **110**.

[0082] In an embodiment, a mounting interface **105** may be releasably attached to a female dovetail mount surface **505** of a magnification ring **501**. In another embodiment, a mounting interface **105** may be permanently attached to a female dovetail mount surface **505** of a magnification ring **501** via one or more adhesives for adhering the set screws **160** and **161** within the threaded holes **507** and **508**. Suitable adhesives for use with set screws **160** and **161** may include, but are not limited to threadlocker adhesives and two-part epoxy compositions. For purposes of this disclosure, when a mounting interface **105** is operably attached to a magnification ring **501**, in an embodiment the mounting interface **105** may be considered as part of the magnification ring **501** whereby attaching a throw lever device **100** to a mounting device **105** may be also be referred to as attaching the throw lever device **100** to the magnification ring **501**. Moreover, in another embodiment a magnification ring **501** may include a male dovetail connector similar as second male dovetail connector **107** provided as original construction of the magnification ring **501** for receiving a female dovetail mount surface **111** of a base member **110**.

[0083] Referring to FIGS. **25** and **26**, in operation when the throw lever device **100** is set to a non-operable position (see FIG. **25**) a biasing force is applied to the plunger **130** via the biasing member **135** (see vector arrow **V1**) directing the beveled surface **132** of the plunger **130** to an abutment position against the second seat **118** of the socket **112** of the base member **110**. As the lever member **120** is directed to an operable position (see directional arrow **A3** in FIG. **26**), the biasing force of the biasing member **135** is overcome by the forced engagement of the beveled surface **132** of the plunger **130** against curved faces **113A** and **114A** (see FIGS. **23** and **24**) of the opposing side wall members **113** and **114** of the socket **112** (see vector arrow **V2**). Once the lever member **120** is directed to an operable position as shown in FIG. **26**, a biasing force is applied to the plunger **130** via the biasing member **135** (see vector arrow **V3**) directing the beveled surface **132** of the plunger

130 to an abutment position against the first seat **117** of the socket **112**.

[0084] The components comprising a throw lever device **100** and mounting interface **105** of this disclosure may be constructed of one or more materials durable for one or more operations and/or as may be required by law or regulation. Suitable materials of construction may include, but are not necessarily limited to those materials resistant to chipping, cracking, excessive bending and reshaping as a result of ozone, weathering, heat, moisture, other outside mechanical and chemical influences, as well as physical impacts. Suitable materials of construction may include one or more metals, plastics, rubbers, woods, filled composite materials, and combinations thereof. Suitable metals may include, but are not necessarily limited to stainless steel, hardened steel, mild steel, aluminum, copper, nickel, brass, titanium, and combinations thereof. Suitable plastics may include, but are not necessarily limited to glass-filled polymers, durable plastic composite materials, and combinations thereof. One suitable glass-filled polymer may include, but is not necessarily limited to glass-filled nylon. One or more adhesives and/or sealants may be used to secure and/or seal one or more components described herein. In addition, one or more heat treatments and/or one or more coatings may be applied to one or more components of a throw lever device **100** and/or mounting interface **105** described herein.

[0085] Referring to FIG. **23**, in one embodiment the boss member **140** and the first extension member **150**, as well as one or more additional extension members, may include corresponding ornamental surface features as shown. In an embodiment, the boss member **140** and the first extension member **150**, as well as one or more additional extension members, may include knurling or other grip enhancing surface configuration(s). In another embodiment, one or more grip materials may be applied to the outer surface of one or more of the boss member **140**, the first extension member **150**, and one or more additional extension members as desired, e.g., one or more rubber grip materials, one or more plastic grip materials, one or more leather grip materials, and combinations thereof. In another embodiment, one or more of the boss member **140**, the first extension member **150**, and one or more additional extension members may include ornamental surface features different than other components of the lever member **120**.

[0086] The disclosure will be better understood with reference to the following non-limiting examples, which are illustrative only and not intended to limit the present disclosure to a particular embodiment.

Example 1

[0087] With reference to FIG. **27**, in a first non-limiting example a throw lever device **100** and mounting interface **105** assembled for use with a magnification ring **501** of an optical sight **500** may comprise the dimensions as listed in Table 1.

TABLE-US-00001 TABLE 1 D1: 4.62 cm (1.82 inches); D2: 1.28 cm (0.50 inches); D3: 1.00 cm (0.39 inches); D4: 3.34 cm (1.31 inches); D5: 2.70 cm (1.06 inches); and D6: 2.62 cm (1.03 inches).

[0088] With reference to FIG. **28**, when the throw lever device **100** and mounting interface **105** of FIG. **27** are mounted to a magnification ring **501** of an optical sight **500**, the throw lever device **100** is spatially disposed adjacent the optical device **500**, at least in part, according to the dimensions as listed in Table 2.

TABLE-US-00002 TABLE 2 D7: 1.19 cm (0.47 inches); and D8: 0.206 cm (0.08 inches).

Example 2

[0089] In a second non-limiting example, an assembly of a throw lever device **100** and a mounting interface **105** is provided as shown in FIGS. **29-37**.

[0090] A system, method, assembly and throw lever device **100** of the present disclosure may be described according to one or more of the following Embodiments.

[0091] Embodiment 1. A system, comprising: [0092] at least a first optical sight comprising a magnification ring; [0093] a device including: [0094] a base member attachable to the magnification ring; and [0095] a lever member pivotally attached to the base member; [0096]

wherein the lever member has a central longitudinal axis; [0097] wherein the lever member comprises a detent assembly axially disposed about the lever member; and [0098] wherein the base member and the detent assembly are operationally configured to retain the lever member in a plurality of fixed positions about the base member.

[0099] Embodiment 2. The system of Embodiment 1, wherein the first optical sight includes an optical axis and the lever member is operationally configured to rotate about a rotational axis in a first direction and an opposite second direction along a plane perpendicular to the optical axis.

[0100] Embodiment 3. The system of Embodiment 2, wherein the plurality of fixed positions include a first fixed position wherein the central longitudinal axis is parallel to the optical axis and a second fixed position offset 90.0 degrees from the first fixed position.

[0101] Embodiment 4. The system of Embodiment 1, wherein the base member includes a socket with a first mating surface for holding the lever member in the first fixed position and a second mating surface for holding the lever member in the second fixed position.

[0102] Embodiment 5. The system of Embodiment 4, wherein the detent assembly of the lever member includes a post member with an eye end pivotally attached to the base member, an annular plunger, a boss member, and at least one biasing member disposed between the annular plunger and the boss member.

[0103] Embodiment 6. The system of Embodiment 5, wherein the detent assembly is operationally configured to bias the annular plunger against the first mating surface and against the second mating surface of the socket.

[0104] Embodiment 7. The system of Embodiment 2, wherein the base member includes a socket operationally configured to limit rotation of the lever member in the first direction and the second direction to 90.0 degrees of rotation.

[0105] Embodiment 8. A system, comprising: [0106] at least a first optical sight and a second optical sight; [0107] at least a first mounting interface and a second mounting interface; [0108] a device including: [0109] a base member; and [0110] a lever member pivotally attached to the base member; [0111] wherein the first optical sight has a first magnification ring with a first mount surface and wherein the second optical sight has a second magnification ring with a second mount surface; [0112] wherein the first mounting interface and the second mounting interface are of a like configuration; [0113] wherein the first mounting interface is releasably attachable to the first mount surface and wherein the second mounting interface is releasably attachable to the second mount surface; [0114] wherein the base member of the device is releasably attachable to the first mounting interface and releasably attachable to the second mounting interface; [0115] wherein the lever member has a central longitudinal axis; and [0116] wherein the lever member comprises a detent assembly axially disposed about the lever member.

[0117] Embodiment 9. The system of Embodiment 8, wherein the base member and the detent assembly are operationally configured to retain the lever member in a plurality of detented positions about the base member.

[0118] Embodiment 10. The system of Embodiment 8, wherein the first magnification ring includes a first outer diameter and the second magnification ring includes a second outer diameter different than the first outer diameter.

[0119] Embodiment 11. The system of Embodiment 8, wherein the detent assembly includes a post member pivotally attached to the base member, an annular plunger, a boss member in threaded communication with the post member, and one or more biasing members, wherein the one or more biasing members are operationally configured to bias the annular plunger against the base member.

[0120] Embodiment 12. The system of Embodiment 11, wherein the base member includes a first mating surface and a second mating surface and wherein the one or more biasing members are operationally configured to bias the annular plunger against the first mating surface and against the second mating surface.

[0121] Embodiment 13. The system of Embodiment 8, wherein each of the first mount surface and

the second mount surface includes a female dovetail mount surface.

[0122] Embodiment 14. The system of Embodiment 13, wherein the first mounting interface includes a first male dovetail connector for mating with the first mount surface and wherein the second mounting interface includes a second male dovetail connector for mating with the second mount surface.

[0123] Embodiment 15. A device for turning a magnification ring of an optical sight, comprising:

[0124] a base member attachable to the magnification ring; and [0125] a lever member pivotally

attached to the base member; [0126] wherein the lever member has a central longitudinal axis; and

[0127] wherein the lever member comprises a detent assembly axially disposed about the lever

member; and [0128] wherein the base member and the detent assembly are operationally

configured to retain the lever member in a plurality of fixed positions about the base member.

[0129] Embodiment 16. An assembly, comprising: [0130] a mounting interface releasably

attachable to a magnification ring of an optical sight; and [0131] a lever assembly including a base

member releasably attachable to the mounting interface and a lever member pivotally attached to

the base member, the lever member comprising a detent assembly axially disposed about the lever

member; [0132] wherein the base member and the detent assembly are operationally configured to

retain the lever member in a plurality of fixed positions about the base member.

[0133] Embodiment 17. An assembly, comprising: [0134] a base member attachable to a

magnification ring of an optical sight; and [0135] a lever member pivotally attached to the base

member, the lever member comprising a detent assembly axially disposed about the lever member;

[0136] wherein the base member and the detent assembly are operationally configured to retain the

lever member in a plurality of fixed positions about the base member.

[0137] Embodiment 18. The assembly of Embodiment 17, wherein the base member is attachable

to the magnification ring via one or more adhesives.

[0138] Embodiment 19. An assembly, comprising: [0139] a mounting interface; [0140] a base

member; and [0141] a lever member; [0142] wherein the mounting interface is releasably

attachable to a magnification ring of an optical sight; [0143] wherein the base member is releasably

attachable to the mounting interface; and [0144] wherein the lever member is pivotally attached to

the base member; [0145] wherein the lever member comprises a detent assembly axially disposed

about the lever member; and [0146] wherein the base member and the detent assembly are

operationally configured to retain the lever member in a plurality of fixed positions about the base

member.

[0147] Embodiment 20. A method for turning a magnification ring of an optical sight, the method

comprising: [0148] manually turning the magnification ring via a device attached to the

magnification ring, the device including: [0149] a base member attachable to the magnification

ring; and [0150] a lever member pivotally attached to the base member; [0151] wherein the lever

member has a central longitudinal axis; and [0152] wherein the lever member comprises a detent

assembly axially disposed about the lever member; and [0153] wherein the base member and the

detent assembly are operationally configured to retain the lever member in a plurality of fixed

positions about the base member including a first fixed position for turning the magnification ring.

[0154] Embodiment 21. A method, comprising the following steps: [0155] adjusting magnification

of a variable optic sight by manually turning a device attached to a magnification ring of the

variable optic sight, the device comprising: [0156] a base member attachable to the magnification

ring; and [0157] a lever member pivotally attached to the base member; [0158] wherein the lever

member has a central longitudinal axis; [0159] wherein the lever member comprises a detent

assembly axially disposed about the lever member; and [0160] wherein the base member and the

detent assembly are operationally configured to retain the lever member in a plurality of fixed

positions about the base member including a first fixed position for turning the magnification ring.

[0161] Embodiment 22. A method, comprising: [0162] turning a magnification ring via an

assembly attached to the magnification ring, the assembly including: [0163] a mounting interface

attachable to the magnification ring; and [0164] a base member attachable to the mounting interface; and [0165] a manually operated lever member pivotally attached to the base member; [0166] wherein the lever member has a central longitudinal axis; [0167] wherein the lever member comprises a detent assembly axially disposed about the lever member; and [0168] wherein the base member and the detent assembly are operationally configured to retain the lever member in a plurality of fixed positions about the base member.

[0169] Embodiment 23. A system, comprising: [0170] at least a first optical sight comprising a magnification ring; [0171] a throw lever device including: [0172] a base member attachable to the magnification ring; and [0173] a lever member pivotally attached to the base member; [0174] wherein the lever member has a central longitudinal axis; [0175] wherein the lever member comprises a detent assembly axially disposed about the lever member; and [0176] wherein the base member and the detent assembly are operationally configured to retain the lever member in a plurality of fixed positions about the base member.

[0177] Embodiment 24. A system, comprising: [0178] at least a first optical sight and a second optical sight; [0179] at least a first mounting interface and a second mounting interface; [0180] a throw lever device including: [0181] a base member; and [0182] a lever member pivotally attached to the base member; [0183] wherein the first optical sight has a first magnification ring with a first mount surface and wherein the second optical sight has a second magnification ring with a second mount surface; [0184] wherein the first mounting interface and the second mounting interface are of a like configuration; [0185] wherein the first mounting interface is releasably attachable to the first mount surface and wherein the second mounting interface is releasably attachable to the second mount surface; [0186] wherein the base member of the throw lever device is releasably attachable to the first mounting interface and releasably attachable to the second mounting interface; [0187] wherein the lever member has a central longitudinal axis; and [0188] wherein the lever member comprises a detent assembly axially disposed about the lever member.

[0189] Embodiment 25. A throw lever device for turning a magnification ring of an optical sight, comprising: [0190] a base member attachable to the magnification ring; and [0191] a lever member pivotally attached to the base member; [0192] wherein the lever member has a central longitudinal axis; and [0193] wherein the lever member comprises a detent assembly axially disposed about the lever member; and [0194] wherein the base member and the detent assembly are operationally configured to retain the lever member in a plurality of fixed positions about the base member.

[0195] Embodiment 26. A throw lever system, comprising: [0196] one or more optical sights, each of the one or more optical sights having a magnification ring with a mount surface; and [0197] a device including a (1) base member releasably attachable to each mount surface of each of the one or more optical sights, and (2) a lever member pivotally attached to the base member; [0198] wherein the lever member is operationally configured to set to at least two fixed positions about the base member.

[0199] Embodiment 27. A method, comprising: [0200] turning a throw lever device of this disclosure when the throw lever device is attached to a magnification ring of an optical sight.

[0201] Although the present disclosure is described above in terms of various exemplary embodiments and implementations, it should be understood that the various features and functionality described in one or more of the individual embodiments are not limited in their applicability to the particular embodiment with which they are described, but instead might be applied, alone or in various combinations, to one or more other embodiments whether or not such embodiments are described and whether or not such features are presented as being a part of a described embodiment. Thus, the breadth and scope of the claimed invention should not be limited by any of the above-described embodiments.

[0202] Terms and phrases used in this disclosure, and variations thereof, unless otherwise expressly stated, should be construed as open-ended as opposed to limiting. As examples of the foregoing: the term “including” should be read as meaning “including, without limitation” or the like, the term

“example” is used to provide exemplary instances of the item in discussion, not an exhaustive or limiting list thereof, the terms “a” or “an” should be read as meaning “at least one,” “one or more,” “one or a plurality” or the like.

[0203] Persons of ordinary skill in the art will recognize that many modifications may be made to the present disclosure without departing from the spirit and scope of the disclosure. The embodiment(s) described herein are meant to be illustrative only and should not be taken as limiting the invention, which is defined in the claims.

Claims

1. A system, comprising: at least a first optical sight comprising a magnification ring; and at least a first device including: a base member; and a lever member pivotally attached to the base member; wherein the magnification ring includes a mount surface; and wherein the base member includes a mounting surface releasably attachable to the mount surface of the magnification ring.
2. The system of claim 1, wherein the mounting surface of the base member includes a planar mounting surface and the mount surface of the magnification ring includes a planar mount surface.
3. The system of claim 1, wherein the mounting surface of the base member includes a non-planar mounting surface and the mount surface of the magnification ring includes one or more conformal contacts for the mounting surface of the base member.
4. The system of claim 1, wherein the lever member comprises a detent assembly axially disposed about the lever member and wherein the base member and the detent assembly are operationally configured to retain the lever member in a plurality of fixed positions about the base member.
5. The system of claim 4, wherein the detent assembly is operationally configured to be biased against different surfaces of the base member to establish the plurality of fixed positions.
6. The system of claim 4, wherein the detent assembly includes a post member, a plunger, a boss member, and one or more biasing members.
7. The system of claim 1, wherein the mount surface of the magnification ring includes one or more threaded holes and the base member includes one or more through holes corresponding to the one or more threaded holes for receiving one or more threaded fasteners.
8. A system, comprising: at least a first optical sight comprising a magnification ring; at least a first device including: a base member; and a lever member pivotally attached to the base member; and at least a first mounting interface; wherein the magnification ring includes a mount surface; and wherein the at least a first mounting interface is releasably attachable to the mount surface of the magnification ring and wherein the base member is releasably attachable to the at least a first mounting interface.
9. The system of claim 8, wherein the mount surface of the magnification ring includes a first slot type female dovetail mount surface, wherein the at least a first mounting interface includes a first male dovetail surface for forming a first dovetail joint with the first slot type female dovetail mount surface, wherein the at least a first mounting interface includes a second male dovetail surface and wherein the base member includes a second slot type female dovetail mount surface for forming a second dovetail joint with the second male dovetail surface.
10. The system of claim 8, wherein the lever member comprises a detent assembly axially disposed about the lever member and wherein the base member and the detent assembly are operationally configured to retain the lever member in a plurality of fixed positions about the base member.
11. The system of claim 9, wherein the lever member comprises a detent assembly axially disposed about the lever member and wherein the base member and the detent assembly are operationally configured to retain the lever member in a plurality of fixed positions about the base member.
12. The system of claim 11, wherein the detent assembly is operationally configured to be biased against different surfaces of the base member to establish the plurality of fixed positions.
13. The system of claim 8, wherein the mount surface of the magnification ring includes one or

more first threaded holes and the at least a first mounting interface includes one or more second threaded holes corresponding to the one or more first threaded holes for receiving one or more first threaded fasteners there through in a manner effective for the one or more first threaded fasteners to mate with the one or more first threaded holes, and wherein the base member includes one or more third threaded holes corresponding to the one or more second threaded holes for receiving one or more second threaded fasteners there through in a manner effective for the one or more second threaded fasteners to mate with the one or more second threaded holes.

14. An assembly for turning a magnification ring of an optical sight, comprising: a base member; a lever member; and a mounting interface; wherein the lever member is attached to the base member; wherein the mounting interface is releasably attachable to a mount surface on the magnification ring of the optical sight; and wherein the base member is releasably attachable to the mounting interface.

15. The assembly of claim 14, wherein the lever member comprises a detent assembly axially disposed about the lever member and wherein the base member and the detent assembly are operationally configured to retain the lever member in a plurality of fixed positions about the base member.

16. The assembly of claim 15, wherein the detent assembly includes a post member, a plunger, a boss member, and one or more biasing members.

17. The assembly of claim 14, wherein the mounting interface includes one or more first threaded holes and the base member includes one or more second threaded holes corresponding to the one or more first threaded holes for receiving one or more threaded fasteners there through in a manner effective for the one or more threaded fasteners to mate with the one or more first threaded holes of the mounting interface.

18. The assembly of claim 14, wherein the lever member is pivotally attached to the base member.
